The problem and promise of modeling plant leafout under warmer winters

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Abstract

How warmer winters affect plant leafout has major implications for carbon storage and ecosystem stability. Recent evidence that the advance of spring budburst has slowed with anthropogenic climate change has reignited a decades-old debate over winter chilling—the concept that woody temperate plants require sufficient winter cool temperatures to budburst each spring. The problem with chilling is that it does not correspond with an observable event. This makes models of it rarely falsifiable and of limited use for advancing scientific understanding. In this paper, we review how current research built on the concept of chilling combined with limited data has led to the proliferation of untestable models that are consistent with a wide range of forecasts and provide little insight into actual plant biology. We argue that the concept of chilling should be reevaluated like past theoretical constructs of limited scientific value (such as 'dormin' or 'anthesin') to help accelerate our understanding of plant development. This can be done by leveraging new insights on the biochemistry and physiology of dormancy to build better models that fit to both experimental and observational data, and by designing better experiments to more rigorously test theory.

Keywords: Chilling, spring phenology, dormancy, endodormancy, process-based modeling, forests

Plant leafout in the spring has shifted weeks earlier in many regions due to warming from anthropogenic climate change with consequences for a wide range of ecosystem services (Keenan *et al.*, 2014; Pörtner *et al.*, 2022). Which underlying processes drive this trend, however, is debated, as recent research suggests winter warming may slow or stall further advances (Fu *et al.*, 2015; Piao *et al.*, 2017). Such reports focus on a two-step model of leafout where plants first require cool winter temperatures, often called 'chilling,' before they can accumulate enough warm temperatures—'forcing'—to leafout each year (Figure 1). This model of chilling is only one of many models proposed since the concept was first introduced (Dennis, 2003; Basler, 2016; Spafford *et al.*, 2025). But, while forcing ends in an observable event such as leafout, the day plants achieve their chilling requirements coincides with no known measurable event.

Despite chilling relating to an unobserved event, decades of work suggest it impacts budburst. Experimental studies, which use controlled temperatures to vary the hypothesized amount of chilling, almost always find evidence that cool temperatures affect the timing and/or success of plant budburst (Erez & Lavee, 1971; Charrier *et al.*, 2015; Baumgarten *et al.*, 2021). Observational studies, which estimate chilling requirements or effects of chilling using observed budburst and temperature data from plants (usually woody plants) in natural settings, report similar effects (Baumgarten *et al.*, 2021). Models generally build on both types of data to predict how winter temperatures impact leafout (Charrier *et al.*, 2015), but new research has highlighted major potential flaws these models.

Current models of chilling predict wildly different leafout times for the same study system and temperature regime (Guy, 2014; Chuine *et al.*, 2016). Even simple models yield radically different predictions for the exact same conditions depending on the calibration data (Olsson & Jönsson, 2014) or which parameters are treated as 'known' (Figure 2). Such extreme variability of spring phenology predicted from current models highlights fundamental gaps in our understanding. At the same time, multiple papers now suggest estimates of 'chilling' could easily be artifacts of poor models or correlated observational climate data (Wolkovich *et al.*, 2021; Gao *et al.*, 2024).

Now appears an opportune time to address the problems in our concept of chilling in models of leafout. Because accurate forecasts of spring phenology are critical for carbon sequestration (Green & Keenan, 2022; Keenan *et al.*, 2014), crop yields and relating food security (Lee *et al.*, 2024; Leisner, 2020), there is widespread interest in improving chilling models (Luedeling *et al.*, 2015; Chuine *et al.*, 2016). At the same time, recent results from molecular and cellular studies of

dormancy are providing insights into when and how chilling works (Pan *et al.*, 2023; Zhu *et al.*, 2021), which could rule in—or out—some of the myriad current models. Here, we review the theory of chilling, its origins and potential problems, as well as new opportunities for major advances and how shifting current practices could accelerate progress. We argue that revisiting the current concept of chilling—and its utility—could help build better models and design better experiments.

What is chilling?

How plants in temperature-limited systems avoid leafout during warm spells in the winter has long been debated by plant biologists (e.g., Lamb, 1948; Weinberger, 1950). Most work to date has focused on the idea that plants enter some form of dormancy in the fall, which is then released before warm temperatures in the spring begin. This idea hypothesizes that the slow accumulation of cool temperatures—or chilling—over the winter extends through periods of short warm spells in the winter and thus prevents leafout before spring.

The term 'chilling' is now used across numerous fields in plant biology to refer to a process where dormant embryonic tissues exposed to cool temperatures accelerate a developmental event—usually budburst—that later occurs after warm temperatures (Fig. 1). The term relates to vernalization—cool temperatures required for flowering (Kim *et al.*, 2009; Kim, 2020)—where much work has focused on the herbaceous species *Arabidopsis thaliana*. Today the term 'chilling' is most often used for dormant buds that later transition to leafout or flowering, but also used for seeds (though 'cold stratification,' which indicates an additional requirement of moist conditions, is more often used for seeds, Baskin & Baskin, 2000; Considine & Considine, 2016). While we also focus on chilling in regards to bud dormancy, the mechanisms underlying chilling requirements of seeds likely provide important parallels to bud dormancy (Leida *et al.*, 2012; Penfield & Springthorpe, 2012; Sato & Yamane, 2024); evolution is more likely to co-opt an existing mechanism than to start de novo.

Much of our fundamental understanding of chilling for buds comes from studies on temperate woody fruit crops where chilling can be critical to yield. Peach trees planted into warmer climates well outside their range often have extremely low fruitset because most flower buds do not burst (Weinberger, 1950; Overcash & Campbell, 1955; Erez *et al.*, 1971). Early studies of this phenomenon with related experiments found cuttings (cut ends of dormant branches from trees) exposed to cool temperatures in controlled chambers (with a standard of 0-7.2°C for many fruit tree studies

Weinberger, 1950) burst more fully and more quickly (Vitasse & Basler, 2014). These studies underlie most of the models of chilling used today for crops and wild tree species (Dennis, 2003).

Focused on how chilling can accelerate observable events, researchers have calculated how much 'chilling' is required for leafout of forest trees from cutting experiments similar to those used for peaches (reviewed in Ettinger *et al.*, 2020), ground observations of budburst (Luedeling *et al.*, 2009b), and satellite measures of greenup (Kaduk & Los, 2011). These estimates rely on phenological models that have become critical across a suite of fields. In climatology and global change biology, modeling how vegetation on the land surface responds to anthropogenic climate change affects spring leafout and carbon storage (Schwartz, 1994; Cleland *et al.*, 2007; Chuine *et al.*, 2013). In crop biology 'chill units' guide growers in which specific cultivar to plant (Figure 3). Alongside these more macro-scale studies of chilling, molecular and biochemical studies have examined variation in hormones, transcripts and other cellular compounds that coincide or precede budburst, generally using controlled temperatures to vary the hypothesized amount of chilling (e.g., Azeez *et al.*, 2021; Cai *et al.*, 2024, and see Fig. 4).

Today these studies have led to over 30 basic models where accumulated chilling releases plants from dormancy (Basler, 2016; Hufkens *et al.*, 2018). Though early debates considered whether plants were truly dormant or only growing slow ('dormancy' or 'rest' versus 'quiescence'; Considine & Considine, 2016). Most research currently assumes a model with two phases of dormancy: (1) endodormany, the period when chilling occurs, and (2) a period between endodormancy but before the observed event, called ecodormancy (Figure 4).

In most models, chilling only accumulates under certain temperatures—traditionally above zero but below 10°C—with certain temperatures being optimal for the most rapid accumulation of 'chill units,' where some unknown sum of chill units breaks endodormancy (Figure 3). Early models introduced 'chill hours' where chilling only accumulated during hours where the temperature was 0-7.2°C (corresponding to the range of 32-45°F developed for peach trees, Weinberger, 1950). After this, the Utah model expanded the range of temperatures to 0-16°C and allowed higher temperatures (>16°C) to negate previous accumulated chilling (Richardson, 1974); such 'chill units' from the Utah model is still widely used, though the term 'chill units' is often used more broadly (as we do here). Since then, more complex models have continually been developed including the Dynamic model, which is broadly similar to the Utah model in its range (0-18°C) and negation, but adds in a response that varies with duration of exposure and can be enhanced depending on

how the temperatures fluctuate (Erez *et al.*, 1987; Erez & Fishman, 1997; Fishman *et al.*, 1987). The Dynamic model introduced the term 'chill portions' and has been widely promoted for fruit tree chilling estimates (Luedeling *et al.*, 2009a, 2021). For many forest trees, which temperatures are most effective at providing chilling is a common question continually addressed in experiments, with different experiments providing different answers (Vitasse *et al.*, 2013; Baumgarten *et al.*, 2021).

Variation in responses to chilling across and within species yields hundreds of variants of the basic models (Basler, 2016; Hufkens *et al.*, 2018). Complexity comes from the hypothesized diversity of chilling temperature thresholds and sums across species and populations. Most assume different species require different sums of chill units, and may have different lower, upper, and optimal chill temperatures. Within species, populations may require different sums of chill units, with populations in more mild climates—where warm interruptions in the winter are more common—requiring more chill units than those in areas with cold winters, where temperatures rarely rise above zero before spring (Campbell & Sugano, 1979; Leinonen, 1996).

The problem with chilling

An unobserved process

Chilling is a latent, unobserved process. Typically, 'chilling' describes the physiological phase (endodormancy) in which a plant experiences environmental temperatures that induce progress towards the next physiological phase (ecodormancy) that ends in budburst (Figure 1). The problem is that the transition from endo- to ecodormancy currently corresponds with no clearly measurable phenomenon, and thus the properties of each period cannot be determined from the observed budburst, even under experimental conditions in which the temperature can be manipulated.

The parameters of a model matching this hypothesized dormancy process are said to be underdetermined or unidentified. For example, consider a simple model in which one would need to estimate the minimum and maximum temperatures that allow chilling to accumulate (two parameters) and the total sum of those temperature units needed to trigger a shift into the next physiological phase (often called 'endodormancy break') for one additional parameter. Models then need to estimate when plants start and stop accumulating (two more parameters). An experiment that raises temperatures and observes an earlier leafout could be explained by more rapid chilling accumulation

leading to earlier endodormancy break, more rapid ecodormancy break, or an almost limitless mix of the two (Figures 1-2).

To address this, models often assign the start date of endodormancy as known (e.g., starting 1 September in the northern hemisphere) and rely on assumptions to set the end date (Figure 2). A common and widely-used assumption, developed by early work on peaches, is that high and rapid budburst (leaf or flower) is evidence that chilling has been met (i.e. endodormancy has ended, Erez & Lavee, 1971, see also Figure 1), but this assumption has rarely been evaluated for other taxa or populations. The approach of assigning some unknown parameters as known has the benefit of avoiding adding more unknown model parameters, but has led researchers to be overly confident in a model of chilling where more is actually unobserved and unknown than acknowledged.

Hidden assumptions in complicated models

More complex models may seem to solve the identification problem, but often come with their own untestable assumptions. Even if we assume that high and rapid percent budburst signals sufficient chilling, most models today include parameters that cannot be uniquely identified with current data, which explains how one model can often predict a wide variety of leafout possibilities (Figure 2). The current tacit approach to these major issues likely has contributed to the expansion of models over the last few decades without any clear advances. While various models have added complexity via the shape of the optimal temperature range for chilling, allowing accumulated chilling to be reduced, shifting the start date of chilling, and/or allowing chilling and forcing to act simultaneously (Hänninen, 1990; Kramer, 1994; Luedeling *et al.*, 2009a; Pope *et al.*, 2014; Gusewell *et al.*, 2017), none of these have swept through the field.

As experiments and models have become more complicated, data have remained relatively uninformative to try to estimate all the complexities the models include. Further, recent models have often relied on even less data (Hänninen *et al.*, 2019). Many current methods use only observational data of the timing of leafout (or flowering) to attempt to estimate a model of chilling for different species or locations and project it forward to understand effects of anthropogenic climate change (Luedeling *et al.*, 2011; Luedeling, 2012; Gao *et al.*, 2024; Fernandez *et al.*, 2022). Perhaps not surprisingly then, which model is deemed best varies strongly by method, underlying data and approach (Caffarra *et al.*, 2011; Olsson & Jönsson, 2014; Basler, 2016; Hufkens *et al.*, 2018), with no clear pattern.

Unclear aims: Prediction versus mechanistic understanding

The flood of current models of budburst with similar levels of predictive accuracy is not necessarily a problem, depending on the aim. Many models of chilling are used to guide crop and tree planting—preventing planting too far outside the range of climates with sufficient chilling for a species, population, or cultivar. For this aim, models need to translate climate into relevant chilling and forcing values and define appropriate thresholds, but how exactly each model defines chilling and forcing is immaterial if the predictions are accurate enough (many models of 'chill units' are thus based simply on average winter conditions). In such cases moving towards machine learning models—which emphasize predictive accuracy often through black box approaches—may provide more practical models that also clearly acknowledge that the underlying process is obscure (Khodadadzadeh *et al.*, 2024; Bauer *et al.*, 2026).

In contrast, if the aim is biological understanding, then the number of highly contrasting but similarly predictive models suggests a problem with the current approach. Models to help elucidate the process that underlies the concept of chilling should represent the current biological understanding. Thus the specification of the model—what parameters it includes, and how well they can be estimated from the data—becomes more important, as does how to evaluate models. Tests that compare models for how accurate they represent our understanding of plant biology, and not only their predictive accuracies across a range of climate regimes, should guide model development. For this aim, new models are unlikely to be enough without new data and approaches that target the major problems facing models of chilling.

Overcoming the chilling problem

Richer, more informative data from molecular studies and other approaches (Fouché *et al.*, 2023; Walde *et al.*, 2024) hold the promise to elevate chilling from an unobserved complex process to something we understand and can robustly forecast. Taking full advantage of this opportunity, however, would benefit from re-examining the concept of chilling across different disciplines to combine observational and experimental evidence, while providing the opportunity to question the future utility of chilling.

Leverage new insights to overcome problems in the simplest models

New research provides hope that the transition from endo- to ecodormancy is measurable. Molecular insights have long contributed to crop and forest tree models of chilling (Chuine & Regniere, 2017). Decades of work on vernalization have outlined the pathways—and genes—that lead to flowering only after winter's cool temperatures in annual/biennial (herbaceous) populations of *Arabidopsis thaliana* (Figure 4, Wilczek *et al.*, 2009; Kim, 2020). Research has linked some of these pathways to similar ones in woody species and have also highlighted the polysaccharide callose (1,3-β-D-glucan) as potentially pivotal for chilling (van der Schoot *et al.*, 2014; Pan *et al.*, 2023).

Multiple studies across multiple species have now shown that (1) lower temperatures appear to degrade callose and (2) the release/loss of callose appears to re-start cell-to-cell communication before budburst (van der Schoot *et al.*, 2014; Pandey *et al.*, 2026). Taken together, these results suggest the loss of callose may be an indicator of endodormancy release.

If callose is a major controller of endodormancy and its release, chilling models could be limited to those that match the idea of callose degradation. This would include models that have a temperature range over which the enzyme(s) degrading callose is active (Figure 4). In contrast, models using simple temperature thresholds (e.g., all hours below -5°C equally allow chilling) would appear less biologically accurate, as enzymes generally do not work over such a wide range of temperatures. Models have begun to introduce the idea of factors or enzymes that may control dormancy or frost hardiness, with multiple studies now focusing on how starch to sugar dynamics may mediate budbreak (Charrier *et al.*, 2018; Sperling *et al.*, 2019). To date, however, most only add these parts as additional complexity onto of existing models (Caspersen *et al.*, 2025; Charrier *et al.*, 2018), which may limit progress (see 'Hidden assumptions in complicated models').

Callose removal itself may also represent only one downstream event of a larger process aligned to chilling. Many other factors, such as phytohormones (e.g., GA and ABA), also often change at the same time (Figure 4, Rinne *et al.*, 2018; Tylewicz *et al.*, 2018; Pan *et al.*, 2023; André *et al.*, 2022). Yet many of these compounds are also integral to other plant development and defense pathways, making it difficult to tie them directly to bud dormancy and budbreak. Increasing work, however, suggests promise through measuring DAMs (dormancy associated MADS-box genes), which are master regulators of various pathways involved in dormancy for both buds and seeds (Leida *et al.*, 2012; Tuan *et al.*, 2017). For example, Zhang *et al.* (2023) recently showed PpDAM6 and

PpMAPK6 may be useful markers for endodormancy maintenance through their effects on ABA—and thus transitions to ecodormancy. Similarly upregulation of 1,3-β-glucanases responsible for plasmodesmata reopening (Rinne *et al.*, 2011; Singh *et al.*, 2018) suggests the 1,3-β-glucanase family could be yet another potential marker for modelling dormancy progression that eventually allows water reactivation into cells and thus growth (Faust *et al.*, 1991; Kalcsits *et al.*, 2009; Walde *et al.*, 2024). Making these new findings most useful to improving models of chilling will require tying these genes more fully to the process and then finding optimum temperature for their expression and bioactivity (see also supplement, 'The search for better indicators').

Other recent insights similarly suggest that most simplified temperature metrics used in many chilling models may not map to mechanistic realities. For example, new work on how slow growth itself may act a 'long-term thermosensor' (Zhao *et al.*, 2020) adds to an increasing number of studies that suggest plants integrate long-term thermo-sensing in the winter alongside responses to short-term temperatures (Antoniou-Kourounioti *et al.*, 2021; Satake *et al.*, 2022). The best models of leafout may thus need to integrate across multiple timescales. While this could easily complicate models already burdened with complexity, insights at the molecular and cellular levels could rule out models by focusing on new experiments and modeling approaches.

Model experimental and observational data together to test assumptions

Research on chilling could accelerate by working across what today are three fairly separated silos: 1) model building for crops and other forecasting needs (usually called process-based models), 2) experiments at the branch or whole-plant level, and 3) molecular research. Currently, crop biologists, phenological process-based modelers of forest trees, molecular biologists, and modelers of related processes (e.g., hardiness) all develop unique and rarely compared models of dormancy and budburst (but see Kovaleski, 2022), highlighting a major problem, but also a major opportunity. Synthesizing and benchmarking models—and their underlying biological understanding of chilling—across the many research fields developing chilling models today would help identify models that are equivalent and/or perform especially poorly, and could begin to discard some models.

Comparing models more directly would hopefully drive new approaches that estimate chilling by fitting experimental and observational data together in one model. This is rarely done, in part because of how differently they may be observed, including the challenging diversity of environmental conditions across these two data types (for example, many experiments apply cold temperatures in the

dark and include extreme temperature differences, while in observational data photoperiod shifts each day and temperatures are more similar) but also because of separate modeling approaches. Researchers almost never fit process-based models to new empirical data; instead they use so-called 'statistical models' that often follow canonical treatment designs (e.g., ANOVA). Statistical models are usually far simpler, and make a suite of unstated assumptions that contradict the current understanding of chilling; for example, most statistical models used assume a linear response to chilling temperatures and durations (Figure 5), whereas all process-based model include an accumulation function, which is inherently non-linear (Wolkovich *et al.*, 2021). Many of the original studies that led to the concept of chilling, however, bridged across observational and experimental studies more often and leveraged datasets that created greater extremes in observational data—by focusing on crops planted well outside their natural range (e.g., peaches in Florida and Israel, Erez & Lavee, 1971; Richardson, 1974). These approaches did this in part by more explicitly mapping from biological processes to statistical models, making it easier to use the models to define new tests to challenge the biological theory.

Experiments bridging across methods may have the greatest opportunity to provide data that would truly challenge current models of chilling. Testing models across large environmental gradients in the field is one of the best ways to find out where models work—and where they fail. For example, Wilczek *et al.* (2009) tested vernalization models by comparing predicted to observed flowering times in a common garden design across Europe and supported the temperature-dependent growth model by testing its predictions of what happens when growth is altered but temperature is held constant (Zhao *et al.*, 2020). Similar examples for challenging other models of chilling date back over 40 years to when many of the models used today were developed (Richardson, 1974; Chuine *et al.*, 2016; Ettinger *et al.*, 2020), but could take place now. Models of chilling can make predictions under lower field chilling then test them using individuals planted beyond the range (either by planting those individuals now or identifying such cases in forestry provenance trials or similar). Process-based modelers could also challenge their models through more dramatic variation in biology; new experiments that include genetic mutants or similar variants (e.g., Azeez *et al.*, 2021; Pan *et al.*, 2023) could help test the effects of genes that putatively influence endo/ecodormancy under a spectrum of natural conditions (Satake *et al.*, 2022).

Is the concept of chilling necessary for scientific progress?

Results from new molecular experiments designed to identify novel ways to directly observe and measure chilling could upend the current concept of chilling. As increasing evidence suggests the concept aligns with multiple shifts at the cellular and molecular level (Figure 4), researchers across disciplines will need to consider how to continue using the term 'chilling.' One approach is to try to align previous and current definitions through new studies, for example, by testing for callose loss (Yu *et al.*, 2024) alongside previously used markers of dormancy shifts—including the often-used bioassay of high and rapid budburst at higher temperatures (indicating endodormancy release) and additional methods, such as weighing flower buds (Chuine *et al.*, 2016) or tracing water reactivation into cells (Faust *et al.*, 1991; Kalcsits *et al.*, 2009; Walde *et al.*, 2024). Another approach is to work towards new terms that more clearly align with new measurements and insights.

Because chilling is an unobserved process, it may link eventually to one measurable compound or regulator (similar to florigen or auxin) or to a more complicated biology, making the term less useful (Aksenova *et al.*, 2006). Current results suggest the latter, and already the term 'chilling' is used to mean different things. For example, chilling sometimes refers to an accumulation process during an unobserved physiological phase (endodormancy), or to the environmental conditions that plants then respond to with an underlying process. In many experiments, the term 'chilling' refers to a treatment where researchers do not know the physiological phase (for example, cuttings or buds from woody plants are often chilled at 5°C for 6 weeks in the dark in a 'chilling' treatment, then transferred to warm 'forcing' conditions, without any tests that could confirm the phase, such as recording percent budburst at high temperatures, or tracking cellular processes, Flynn & Wolkovich, 2018; Ettinger *et al.*, 2020).

Diverging definitions suggest that the best path for chilling may be to follow that of other terms modern plant biologists may not remember, such as dormin or anthesin (hypothesized compounds to induce dormancy and floral anthesis, respectively, Aksenova *et al.*, 2006; Dörffling, 2015), and let the term fade into obsolescence. Fundamental research on the physiological and molecular biological mechanisms of bud dormancy is primed to discover new measurable processes and observable events related to what we call 'chilling' today. With this, new terms and concepts that better match our improved understanding could emerge to aid research across the disciplines studying chilling. Models for forecasting and new experiments at the branch and whole plant level could focus on predicting and measuring these new events and processes. 'Chill units' for widely planted forest trees

and specific varieties of crops, which are already less linked to the mechanism of chilling, would likely hold on longer. They too, however, may shift to new estimates or units based on better measures, especially as increasing anthropogenic climate change makes getting the mechanistic biology of plant development right all the more important.

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2 Figures

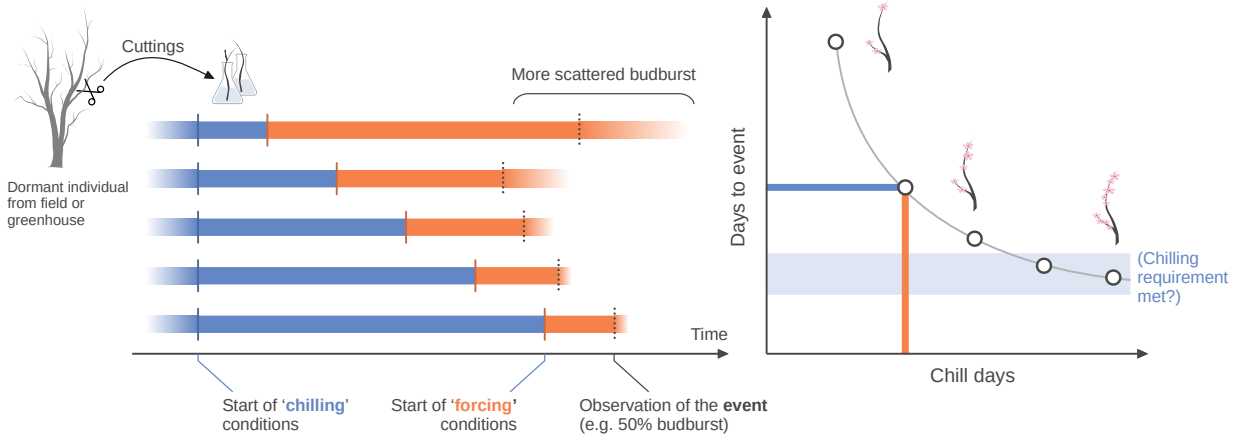


Figure 1: The term chilling generally refers to a process where dormant embryonic tissues exposed to cool temperatures accelerate a plant event that later occurs. While chilling cannot be directly observed currently, experiments for over a hundred years have tried to identify how much chilling plants need by finding the conditions that lead to rapid and full budburst (see right side where chilling is shown as the modeled unit of 'chill days'). Many studies have measured this via experiments that take dormant tissue (e.g., cut ends of tree branches taken from the field during the winter), which they then expose to sequential cool temperature treatments (generally called 'chilling' treatments or conditions) followed by warmer temperature treatments (called 'forcing' treatments or conditions), and observe budburst (left). Required chilling is then defined from these experiments by when additional cool temperatures do not yield faster and more complete budburst (right).

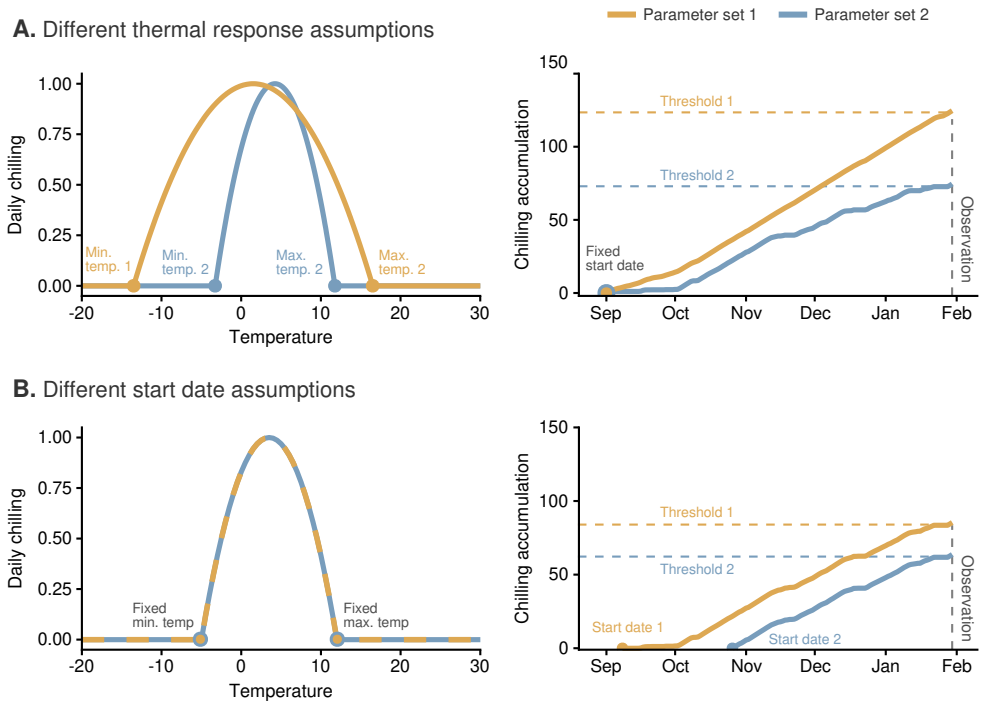


Figure 2: **Current models of chilling cannot uniquely estimate the most important aspects of chilling.** We illustrate this using a simple chilling model with a symmetrical thermal response curve, assuming the end of the chilling phase (often called endodormancy break) is observed (dashed vertical lines in right panes). In both cases, multiple parameter sets reproduce the same observation equally well. **(A)** Estimating the thermal response curve parameters (min. and max. temperatures) and the chilling threshold jointly (with a fixed start date) leads to many potential solutions. For example, a wider temperature range requires a higher threshold to reach the observed date (yellow), while a narrower range requires a lower threshold (blue). **(B)** The same issue happens when the response curve is fixed but the start date and threshold are estimated jointly. An earlier start date with a higher threshold (orange) leads to the same observed date as a later start date with a lower threshold (blue). These two simple examples illustrate that the parameter space contains multiple indistinguishable solutions, even when the end of the chilling phase is known (which is rare, more often models are built with only data on the end of the next phase—observed via budbreak). This issue would be even worse if the thermal response curve, the threshold, and the start date were all estimated jointly—and when the end of the phase is not directly observed. This reality is present in every model of chilling, but rarely discussed and often not even mentioned.

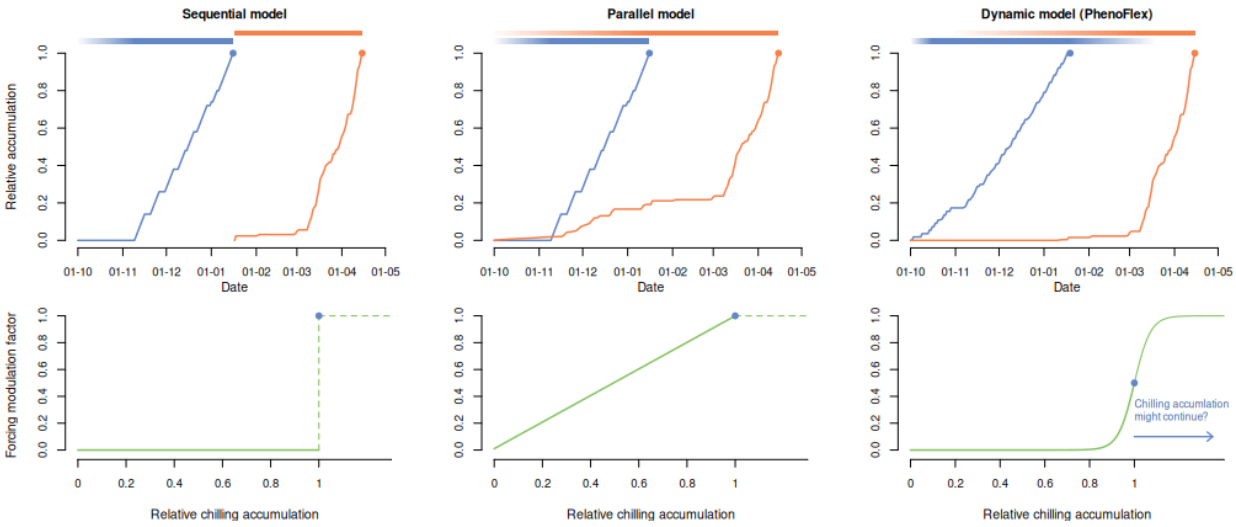
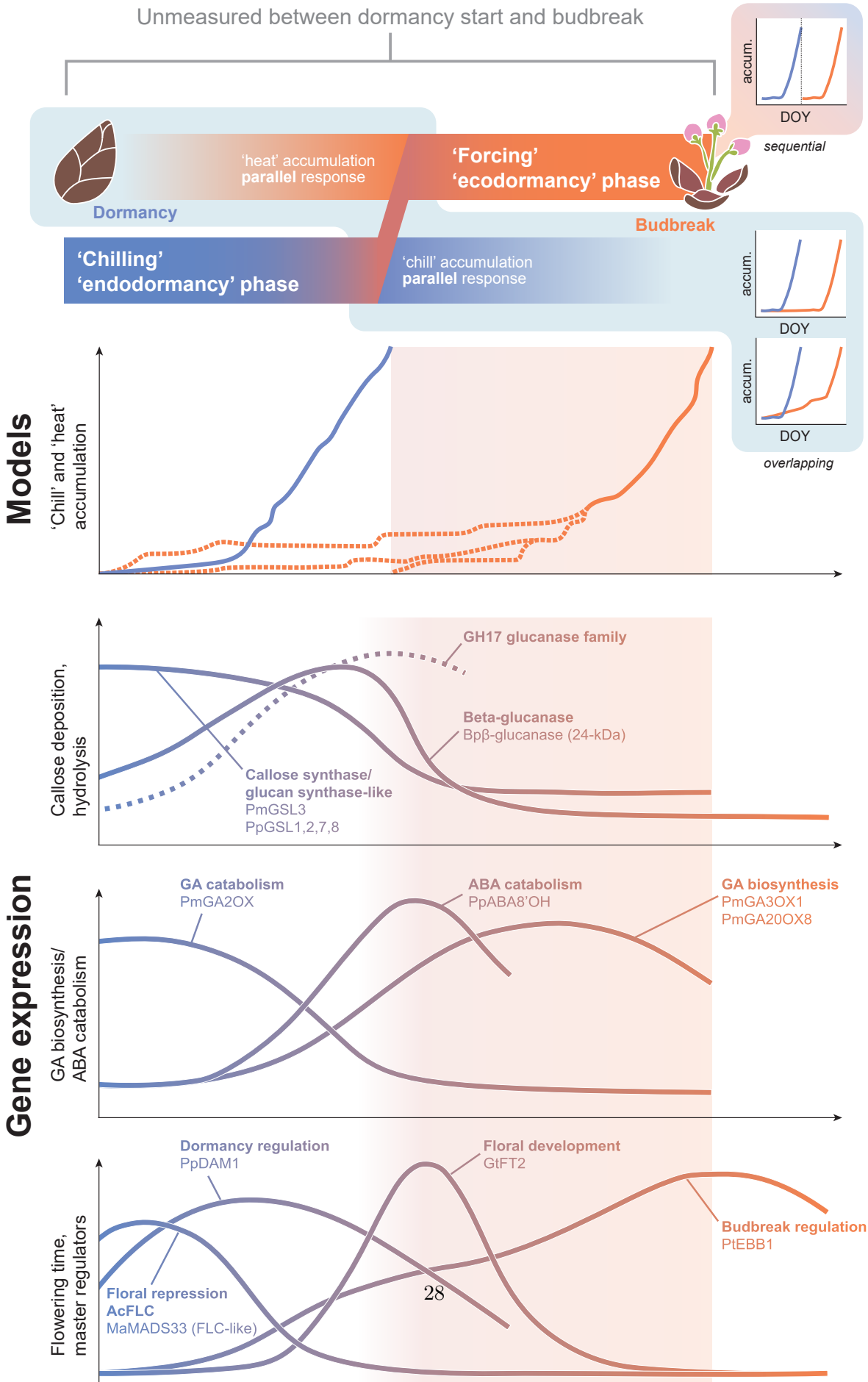


Figure 3: There are a lot of chilling models ... .


Unmeasured between dormancy start and budbreak
accum.
DOY
sequential
'heat' accumulation
parallel response
'Forcing'
'ecodormancy' phase
Dormancy
Budbreak
'Chilling'
'endodormancy' phase
'chill' accumulation
parallel response
accum.
DOY
accum.
DOY
overlapping
Models
'Chill' and 'heat'
accumulation
Callose deposition,
hydrolysis
GH17 glucanase family
Beta-glucanase
Bpβ-glucanase (24-kDa)
Callose synthase/
glucan synthase-like
PmGSL3
PpGSL1,2,7,8
Gene expression
GA biosynthesis/
ABA catabolism
GA catabolism
PmGA2OX
ABA catabolism
PpABA8'OH
GA biosynthesis
PmGA3OX1
PmGA20OX8
Flowering time,
master regulators
Dormancy regulation
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Floral development
GtFT2
Budbreak regulation
PtEBB1
Floral repression
AcFLC
MaMADS33 (FLC-like)

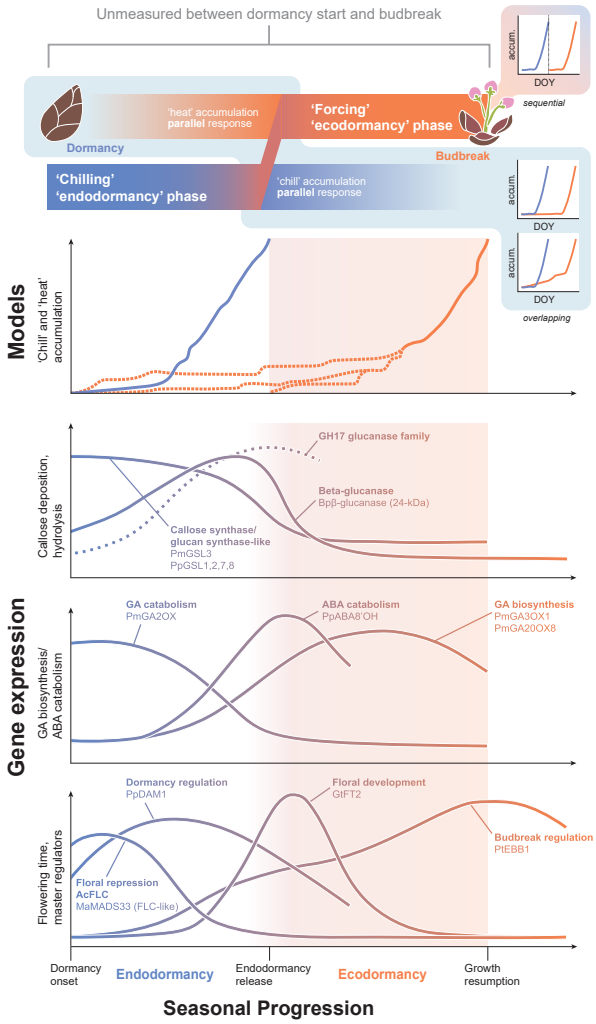


Figure 4: Aligning phenological models and molecular findings. Current phenological models of budbreak (top panel) assume two underlying states that lead to the observed process of budbreak: (1) endodormancy during which plants accumulate sufficient chilling to break endodormancy and transition (shown in blue) to (2) ecodormancy, a period during which plants accumulate sufficient heat or 'forcing' (warm temperatures, shown in orange) to break bud (flower or leaf), phases which may be sequential (darker shading) or occur in parallel (dark and light shading) yielding many variants of this model (second panel, and see Figure 3). Cellular and molecular studies have found a suite of gene expression patterns than may underlie this transition, including shifts in callose (third panel), phytohormones (fourth panel) and those related to flowering time and master regulators (bottom). We show the glucanase family as dotted, given how many there are and that different glucanases are upregulated depending on the dormancy phase (Rinne *et al.*, 2011). We provide information on the species and citations underlying this figure in the Supplement (see 'Additional details on Figure 4')

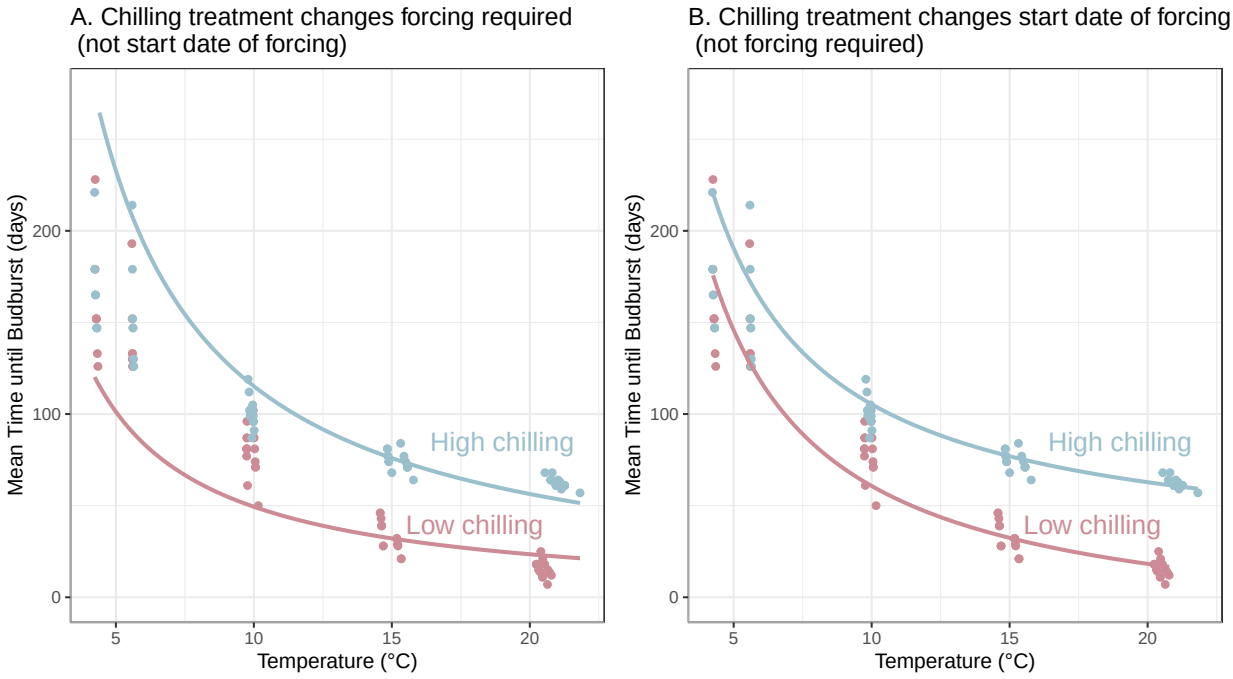


Figure 5: Many experiments studying chilling alter 'chilling' and 'forcing' by varying the duration that cool temperatures are applied ('chilling') followed by different warm temperatures ('forcing'). Fitting simple linear models to these data (e.g., ANOVA models, with main effects of 'chilling' and 'forcing' treatments and their interaction—'chilling' × 'forcing'—to find a sub-additive effect of the two) usually does not match the current biological understanding of chilling. Fitting models that match the proposed accumulation process is possible, however, as we show here using data from Walde *et al.* (2022) for *Quercus robur*, and considering two alternative biologies: (A) cool ('chilling') and warm treatments (x axis) interact, such that longer cool temperatures mean more warm temperatures are required for leafout, versus (B) longer cool temperatures change the start date of the accumulation of warm ('forcing') temperatures (also known as endodormancy break).